

3D CFD Simulation of n-Heptane/Air Combustion in a Combustor

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1 Description of the problem

This assignment focuses on the 3-D simulation of the combustion between air and *n* – Heptane C_7H_{16} inside a 3D combustor. Regarding the geometry of the combustor, the mesh is generated from the provided STL file. Air enters in the domain only through the inlet patch, while n-Heptane is injected in the form of Lagrangian Particle Tracking (LPT). Once they react, the combustion is achieved. To initialize the flow, a precursor simulation is computed, with combustion, chemical reactions and spray deactivated. the base setup is obtained from both Assignment 3 (without considering surface film modeling) and Assignment 4.

2 Setup

2.1 Mesh

The computational mesh is generated from the provided STL file of the 3D combustor. A constraint is imposed on the mesh size, which must not exceed a maximum of 380,000 cells due to computational limitations.

Since the original STL geometry is defined in millimetres, it must first be scaled to metres. This is accomplished using the `surfaceTransformPoints` utility, which generates the updated `combustor.stl` file with dimensions scaled by a factor of `scale (1e-3 1e-3 1e-3)`.

The background mesh is then created using `blockMesh`, with a grid resolution of `(10 33 10)` specified in the `blockMeshDict`. These values are chosen to maintain cells as close to cubic as possible. The extent of the box along the *y*-axis is slightly reduced compared to the original STL geometry in order to avoid open surfaces, since the STL is not closed at the two ends of the duct. This reduction is negligible and does not affect the accuracy of the simulation. The resulting background mesh is shown in Figure 1, alongside the original STL geometry for comparison.

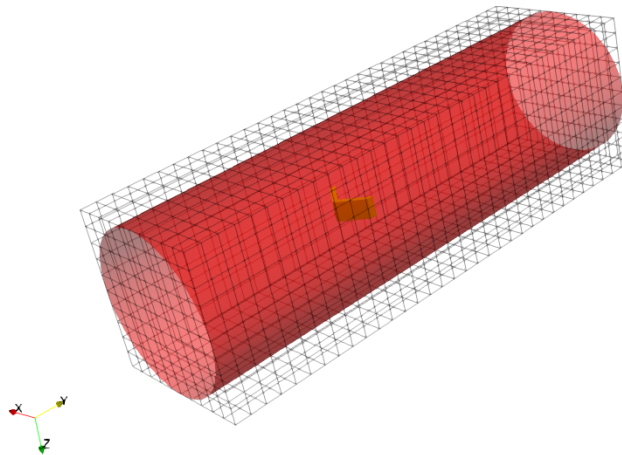


Figure 1: Background mesh and combustor geometry

Once the background mesh is initialized, the refinement is carried out using `snappyHexMesh`. The refinement region, defined through the `refinementBox` in the `snappyHexMeshDict`, is set to include

the V-shaped splitter and to guarantee a higher mesh resolution in areas where the flow is complex and combustion takes place.

The **snappyHexMeshDict** follows the standard three-stage workflow: first, the **castellatedMesh** phase refines the cells and captures geometric features; next, the **snap** phase projects the mesh onto the surface geometry and smooths the boundaries; finally, the **addLayers** phase inserts boundary layer cells near the walls to improve near-wall flow resolution.

The first step is the **castellatedMesh** generation, where all the unwanted cells from the background mesh are removed, as the outer walls and the cells inside the splitter. This as the aim of this project is to model the combustion inside the geometry. In ?? the parameters adopted inside **snappyHexMeshDict** for the **castellatedMesh** are reported.

The second step is the **snap** phase. Here the default parameters taken from the test case (**ahmedBody**) are used.

Finally, the last step is the **addLayers** phase. Absolute sizes are used, and the refinement is concentrated in the **splitter** region, with 4 layers applied. The expansion ratio is set to 1.15, with a first layer thickness of 0.8 mm and a minimum overall thickness of 0.01 mm. These parameters are chosen to ensure a smooth growth of the boundary layer mesh and to capture the near-wall gradients with sufficient accuracy. After some iterations, the combustor mesh is computed with the following characteristics.

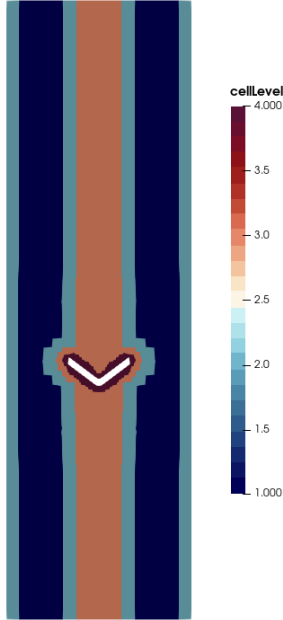


Figure 2: Refinement levels obtained on a slice perpendicular to z -axis

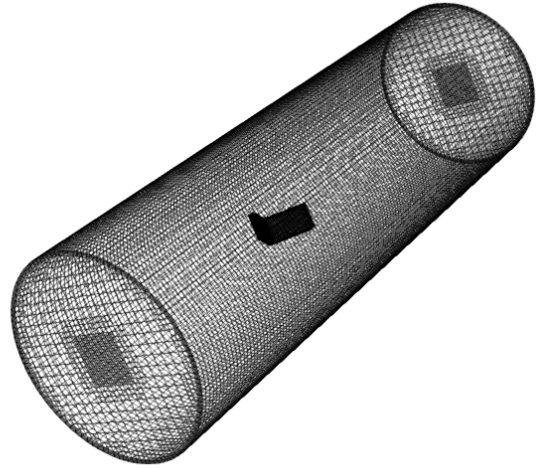


Figure 3: Final Mesh

Parameter	Value
maxLocalCells	375000
maxGlobalCells	300000
minRefinementCells	10
nCellsBetweenLayers	2
Combustor refinement level	(0 0)
Splitter refinement Level	(4 4)
Outer Walls refinement level	(2 2)
Refinement Box level	3

Table 1: Castellated mesh parameters

Parameter	Value
Cells	264520
Non-orthogonality	Max: 64.81 Average: 8.66
Max skewness	0.976436
Max aspect ratio	8.47982

Table 2: Final mesh parameters

2.2 Hot Run Configuration

The main parameters of the simulation setup are summarized in Table 3 and ??. The simulation employs the PaSR (Partially-Stirred Reactor) model to account for the interaction between chemical reactions and turbulence. In this approach, each computational cell is conceptually divided into two locally uniform regions: one where chemical reactions occur, and one where only mixing takes place. The final species concentration is then computed based on mass exchange between these zones, driven by turbulent mixing, providing a more realistic representation of the combustion process.

The chemistry solver **seulex** is selected for its robustness in handling stiff chemical kinetics. The injection of **n-Heptane** is defined in the **cloudProperties** file using a half-sine profile. An additional point (0.2, 0) is added to the profile to prevent negative values during interpolation, ensuring numerical stability.

The governing equations in the domain are solved using the **foamRun** application, with solver **multicomponentFluid**. Key numerical parameters defined in the **controlDict** file include **endTime** = 0.08, and **deltaT** = 1×10^{-6} . Output is written at intervals of **writeInterval** = 0.01. The maximum Courant number (**maxCo**) is set to 0.3 to properly resolve combustion and turbulence dynamics while avoiding numerical instabilities during the transient simulation. The solver operates under the PIMPLE algorithm with 2 correctors, 3 outer correctors and 2 non-orthogonal correctors to ensure convergence and robustness. The relaxation factors for each variable are: $p = 0.3$, $U = 0.7$, $(k|h|\epsilon) = 0.5$. In Table 3 the updated solvers setup is reported.

Table 3: Solver configuration for each field (as defined in **fvSolution**).

Field	Solver	Preconditioner / Smoother	Tolerance	relTol
rho.*	diagonal	-	1e-09	0
p	GAMG	smoother: GaussSeidel	1e-06	0.1
pFinal	GAMG	smoother: GaussSeidel	1e-06	0
(U, k, ϵ , h)	PBiCGStab	DILU	1e-06	0
(U, k, ϵ , h)Final	PBiCGStab	DILU	1e-06	0
(Yi, O2, N2, H2O).*	PBiCGStab	DILU	1e-06	0

Folder	constant
chemistryProperties	chemistry on absTol 1e-8 relTol 0.1 solver seulex
combustionProperties	combustion on model PaSR
physicalProperties	thermoType hePsiThermo
cloudProperties	cloud reactingCloud breakup ReitzDiwakar position (0 -0.285 0) direction (0 1 0) SOI 0 massTotal 10e-5 duration 0.1

Table 4: **constant** folder main parameters

2.3 Cold Run Configuration

A cold run is performed to initiate the flow, with a simulation `endTime` of 0.05 s. The output of this simulation will be the initial condition for the hot run describe in the previous section. To initiate the flow, `.chemistry`, `combustion` and `injection` models are deactivated, and the solver is set to `fluid`, as there are no multi-component simulations. Since combustion is off and flow is affected only by the splitter presence, the `maxCo` is set to 5, in order to reduce the simulation time. For the same reasons, (U, k, ϵ, h) can be solved simply with `smoothSolver` and `deltaT` can be increased up to $1e - 05$.

The boundary conditions used for this simulation are reported in Table 5.

Table 5: Boundary conditions applied at each patch for main variables.

Variable	Inlet	Outlet	outerWalls - splitter.*
U	fixedValue (0 11 0)	zeroGradient	noSlip
p	zeroGradient	waveTransmissive	zeroGradient
T	fixedValue (900 K)	zeroGradient	zeroGradient
k	turbulentIntensityKineticEnergyInlet	zeroGradient	kqRWallFunction
ϵ	turbulentMixingLengthDissipationRateInlet	zeroGradient	epsilonWallFunction
α_t	calculated	calculated	alphatWallFunction
ν_t	calculated	calculated	nutkWallFunction
O ₂	fixedValue (0.233)	inletOutlet	zeroGradient
N ₂	fixedValue (0.766)	inletOutlet	zeroGradient
Ydefault	inletOutlet	inletOutlet	zeroGradient

2.4 Post-Processing

The following post-processing operations are included in the `controlDict` configuration:

- **Residuals:** Computed for pressure, velocity components, enthalpy, turbulent kinetic energy, and dissipation rate to monitor numerical convergence.
- **Qdot:** The volumetric heat release rate is calculated and made available as a scalar field, useful for identifying reaction zones.
- **Vorticity:** The vorticity magnitude field is generated to visualize rotational structures within the flow.
- **Mach Number:** Computed across the entire domain to assess local compressibility effects.
- **Patch Averages:** Mean pressure and temperature values are extracted at both the `inlet` and `outlet` patches, which are later used to analyze overall pressure and thermal performance.
- **Transversal Lines:** Two line probes are defined:
 - At $y = 110$ mm (line2), to analyze species distribution near the upper combustion region;
 - At $y = -90$ mm (line1), located near the lower region.

Each line samples 1000 points along the x-axis, and tracks species such as N₂, O₂, C₇H₁₆, CO₂, and H₂O.

- **Normal Planes:** Multiple cut planes are defined (see ??):
 - `zPlane`: Slice along XY plane at $z = 0$;
 - `yPlane1` and `yPlane2`: Slices along XZ at $y = -90$ mm and $y = 110$ mm;
 - `xPlane`: Slice along YZ at $x = 0$.

These planes export fields in VTK format for post-visualization (p, U, T, O₂, N₂, C₇H₁₆, CO₂, H₂O, Ma, vorticity, Qdot).

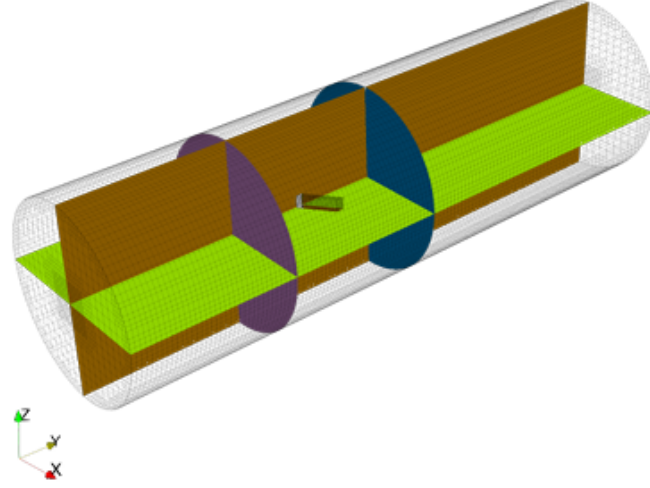


Figure 4: Normal planes adopted for Post-Processing

3 Results

First the results of the precursor run are presented, since they will be used as initial conditions for the hot run simulation.

3.1 Cold Run Results

The results of the precursor simulation in terms of pressure, temperature, and velocity fields are shown in Figure 5. In the pressure field, a clear region of overpressure is observed upstream of the splitter, where it directly interacts with the incoming flow, while a low-pressure wake forms downstream, as expected. The temperature field highlights elevated values near the walls, along with a distinct upward convection pattern, mainly driven by the geometry of the splitter. In the velocity field, a symmetric recirculation region forms behind the splitter. This structure is particularly important, as it promotes mixing and will play a key role during the combustion phase.

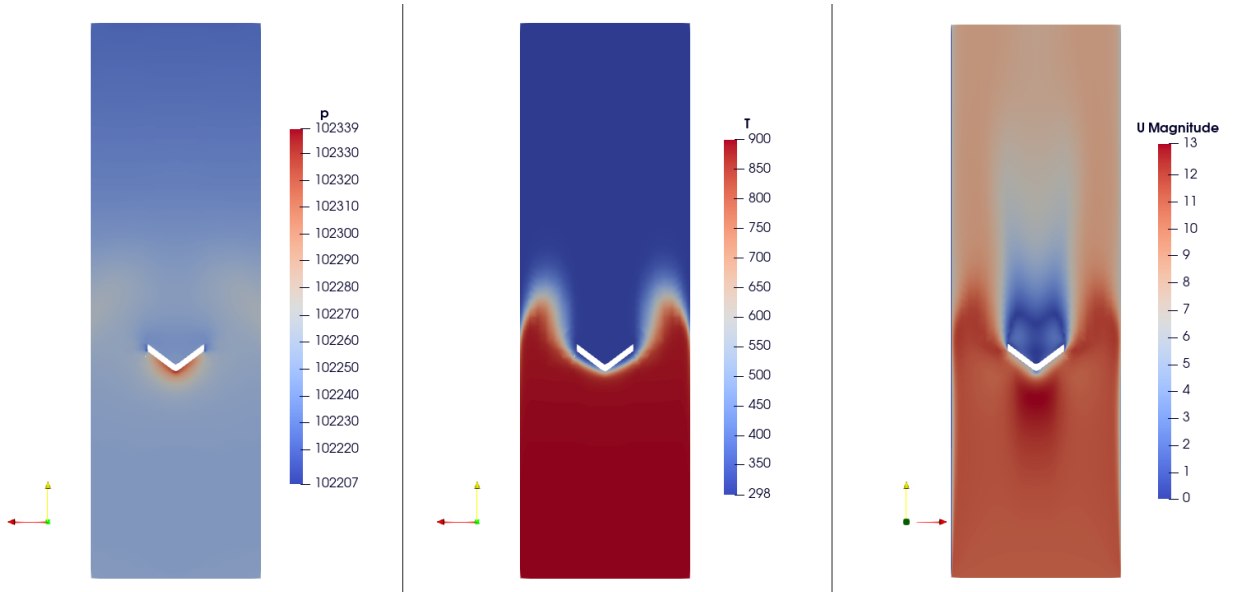


Figure 5: Pressure, temperature and velocity distribution along z-plane

These results are used as initial conditions for the subsequent hot run, where spray injection, chemistry and combustion are activated.

3.2 Hot Run Results

3.2.1 Residuals

The residuals are shown in Figure 6. Among all the variables, pressure exhibits the highest residuals and the most significant oscillations. This behavior is expected, as the simulation involves a compressible flow coupled with a combustion process, both of which introduce strong nonlinearities and numerical stiffness in the pressure field.

For all the other residuals a common trend can be individuated. There is an initial spike during the early iterations, indicating a rapid adjustment of the solution towards a physically meaningful state. Then they display a general decay until they progressively converge towards 10^{-4} .

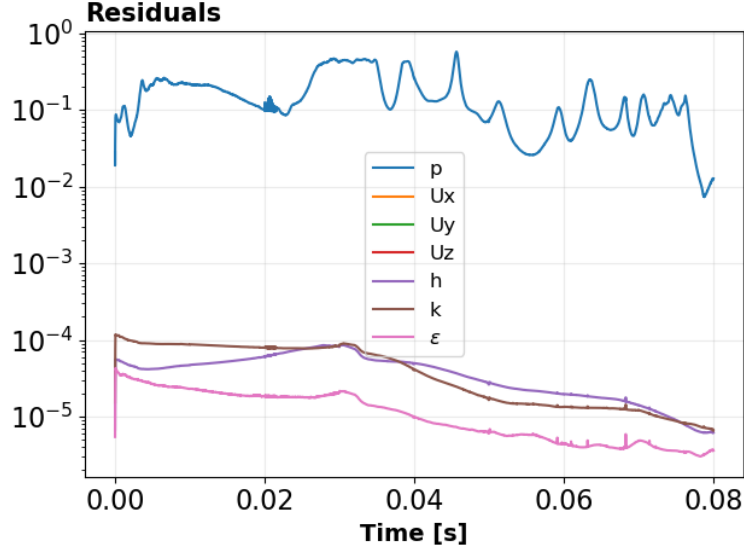


Figure 6: Residuals over time

3.2.2 Temperature Field Distribution

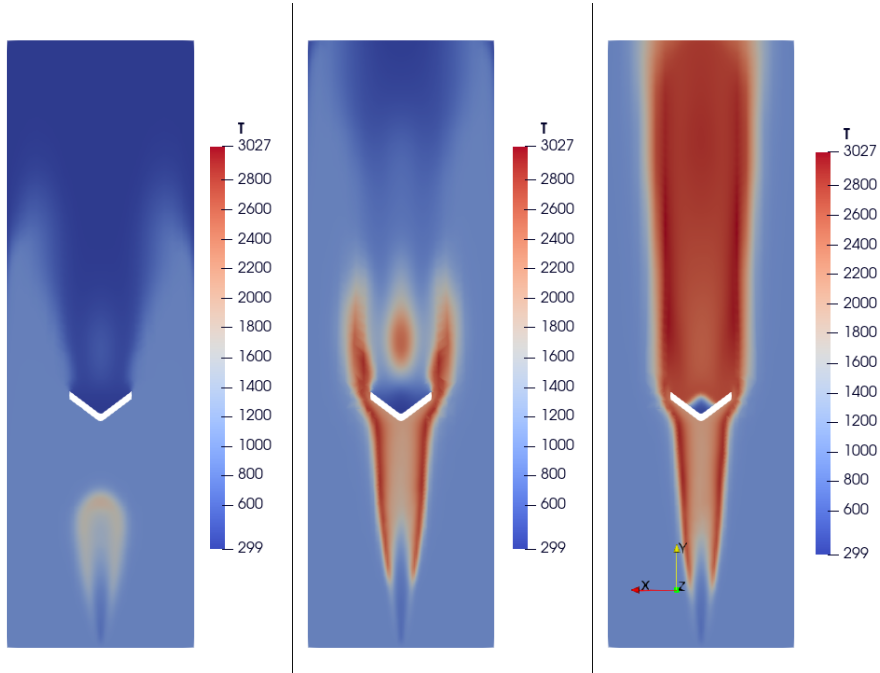


Figure 7: Temperature field distribution along z-plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

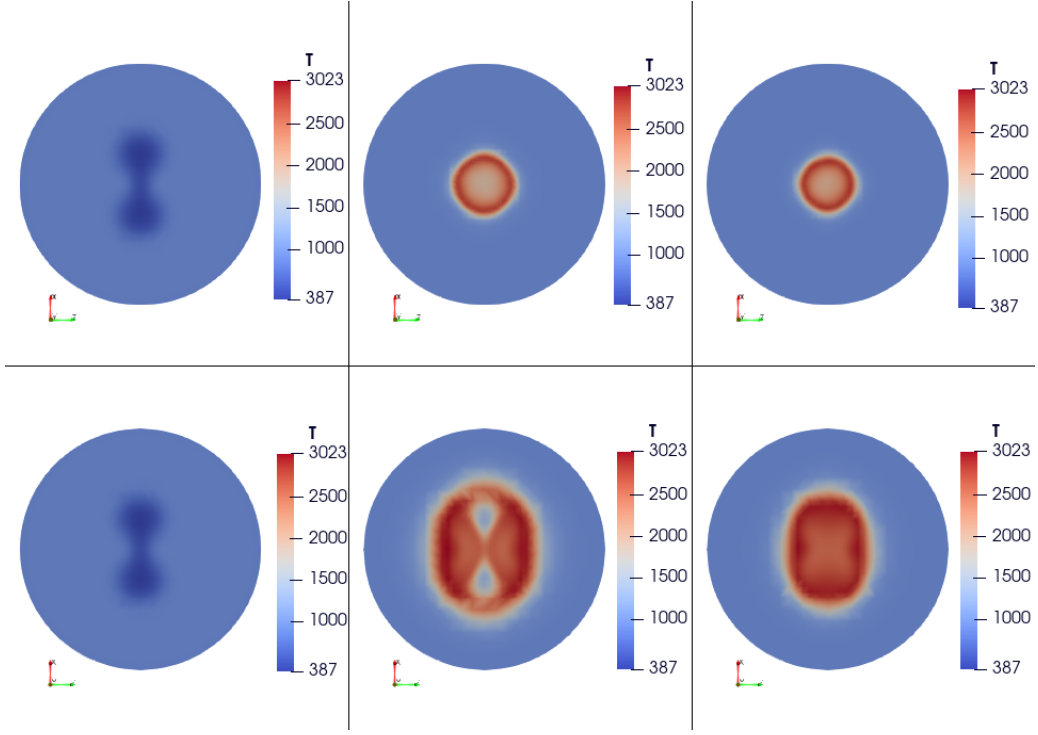


Figure 8: Temperature field distribution along y -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

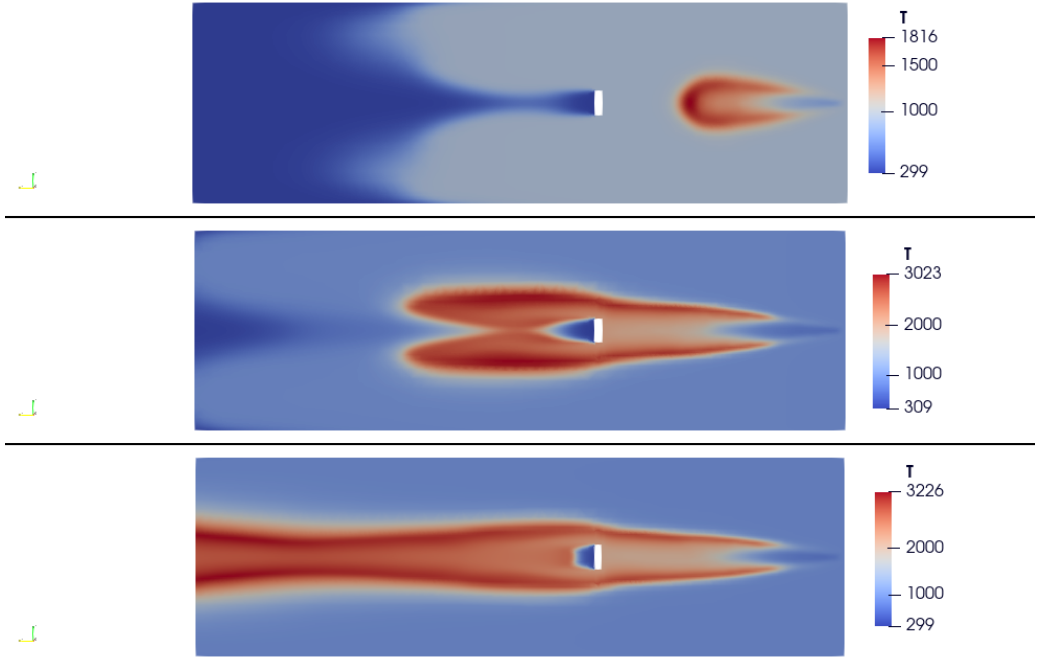


Figure 9: Temperature field distribution along x -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

The temperature contours at $t = 0.02$ s, $t = 0.05$ s, and $t = 0.08$ s are shown in Figure 7, Figure 8 and Figure 9. The temperature evolution throughout the simulation reveals the formation of high-temperature regions along the central axis, particularly downstream of the splitter, where combustion is initiated. As time progresses, enhanced mixing becomes evident, with hot zones expanding radially and interacting with adjacent cooler areas. This dynamic leads to a progressively more uniform temperature distribution by $t = 0.08s$. The V-shaped component plays a key role in directing thermal energy upward while simultaneously promoting symmetry and enhanced mixing through the establishment of recirculation zones near the side walls.

3.2.3 Heat Field Distribution

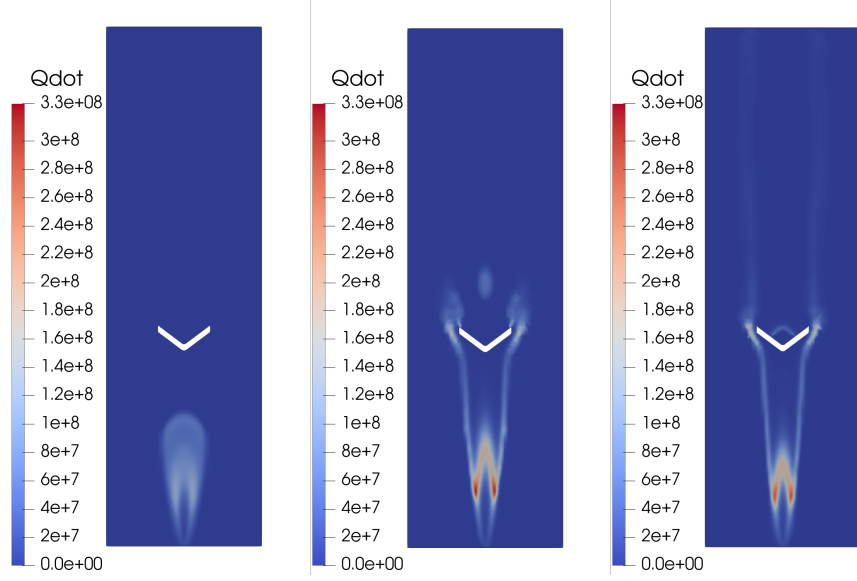


Figure 10: Heat release field distribution along z -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

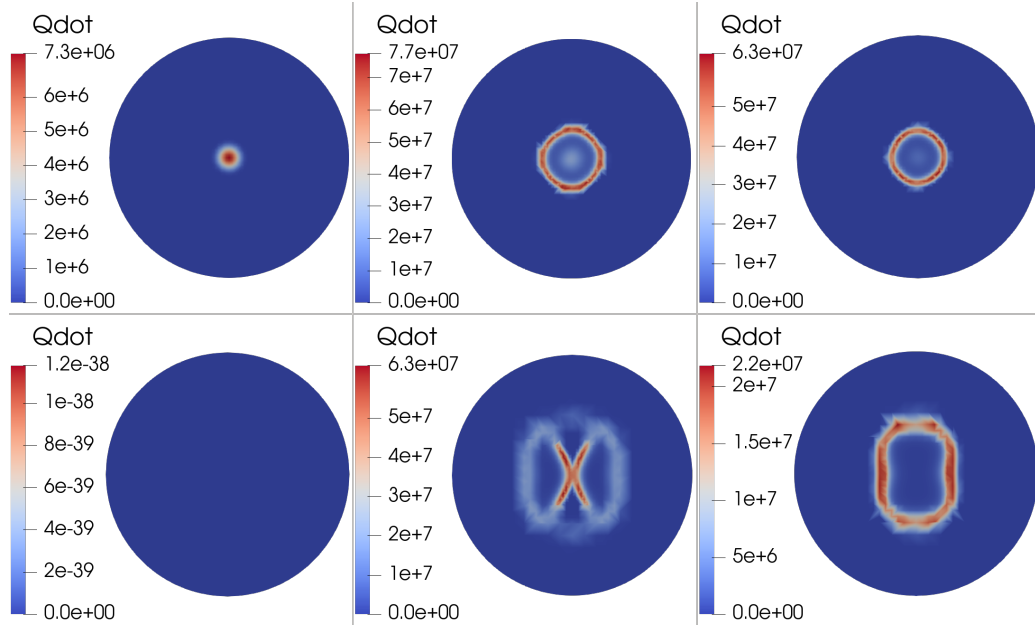


Figure 11: Heat release field distribution along y -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

The evolution of $\dot{Q}$ follows the same trend described for the temperature field in the previous section. The highest values are observed in the upstream region of the combustor, where combustion reaches its peak efficiency due to optimal mixing and ignition conditions. At later time steps, the heat release rate decreases as a consequence of the progressive consumption of fuel and the reduced availability of reactants in the core of the flow.

By looking at Figure 11 can be noticed that $\dot{Q}$ initially appears as a localized spot near the centerline of the combustor. Over time, the $\dot{Q}$ distribution expands radially, forming a ring-like structure that becomes increasingly larger and more diffused. This radial growth reflects the propagation of the flame front and the progressive involvement of outer flow regions in the combustion process. By the final time step, the heat release extends significantly towards the side walls, indicating enhanced mixing and combustion across a wider cross-sectional area

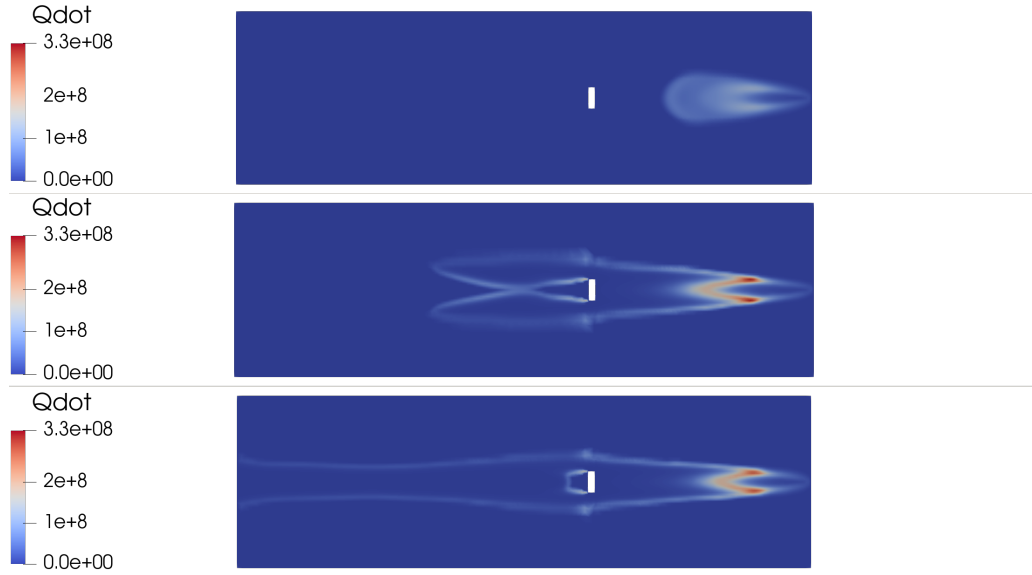


Figure 12: Heat release field distribution along x -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

3.2.4 Pressure Field Distribution

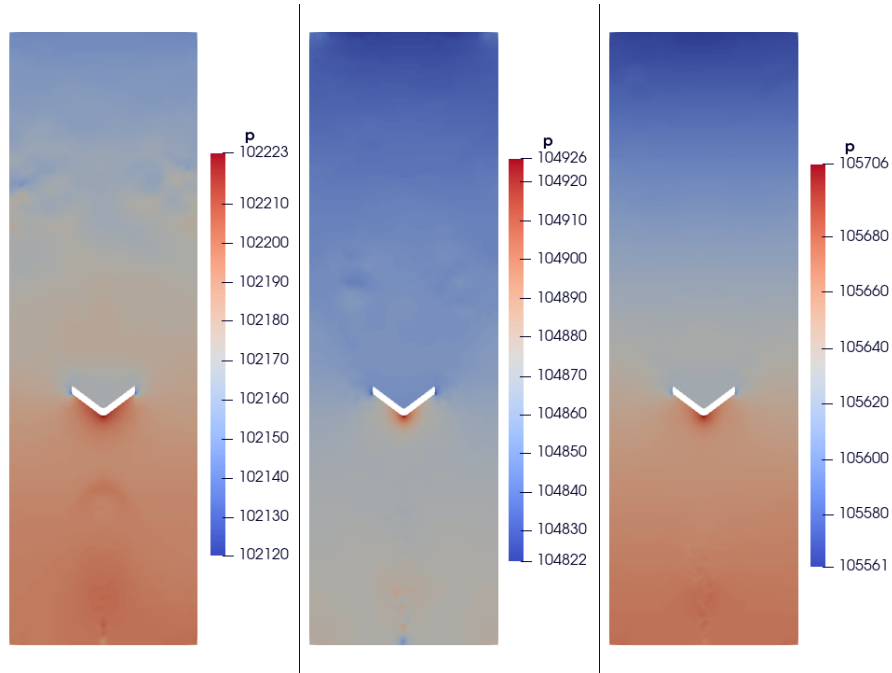


Figure 13: Pressure field distribution along z -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

The evolution of the pressure field is shown in Figure 13 and Figure 14. As expected, a strong compression region develops upstream of the splitter, while a low-pressure zone forms downstream. This under-pressure region results from flow separation and the onset of turbulence caused by the geometry of the splitter. The resulting adverse pressure gradient promotes the formation of recirculation zones, which enhance mixing and contribute to overall flame stabilization.

Furthermore, the view on the x -plane clearly highlights the symmetry of the pressure field around the obstacle. This symmetry confirms the balanced inflow conditions and the effectiveness of the V-shaped geometry in distributing the flow evenly on both sides, despite the presence of strong gradients and complex interactions.

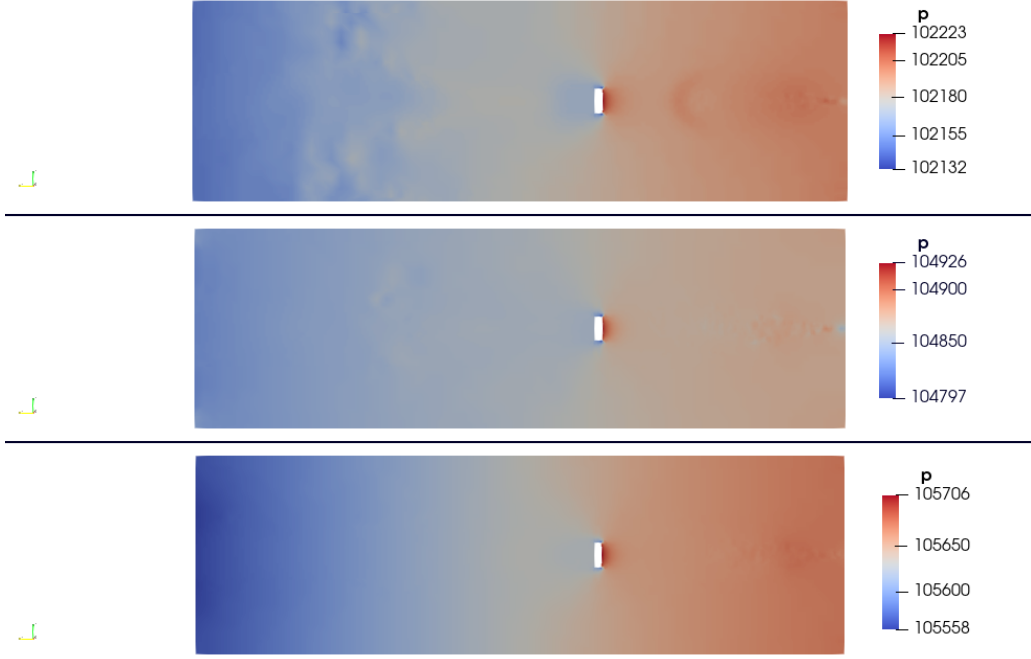


Figure 14: Pressure field distribution along x -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

3.2.5 Velocity and Mach Field Distribution

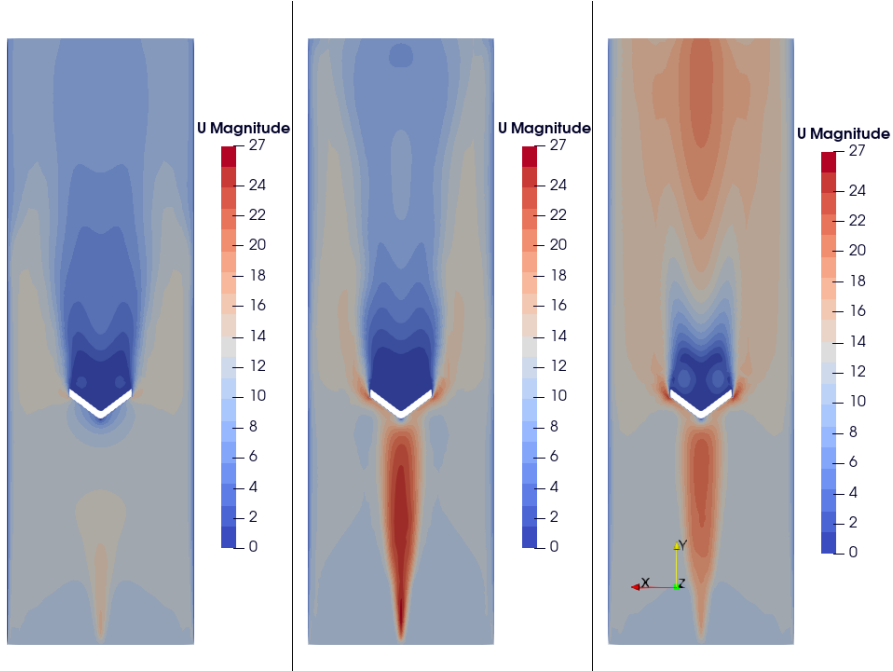


Figure 15: Velocity field distribution along z -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

The velocity field evolution is presented in Figure 15, Figure 16, and Figure 17. High-speed regions are primarily concentrated along the central axis of the combustor, where fuel injection occurs and combustion is initiated. The highest velocities are observed at $t = 0.05s$, coinciding with the peak heat release. Combustion promotes thermal expansion and accelerates the flow, whereas the subsequent reduction in fuel availability leads to a gradual decrease in velocity.

The V-shaped splitter plays a crucial role in organizing the incoming streams, promoting mixing and enhancing symmetry in the velocity field. Over time, the effect of the splitter becomes more evident, with the flow gradually becoming more homogeneous downstream due to enhanced turbulent

mixing and reduced velocity gradients.

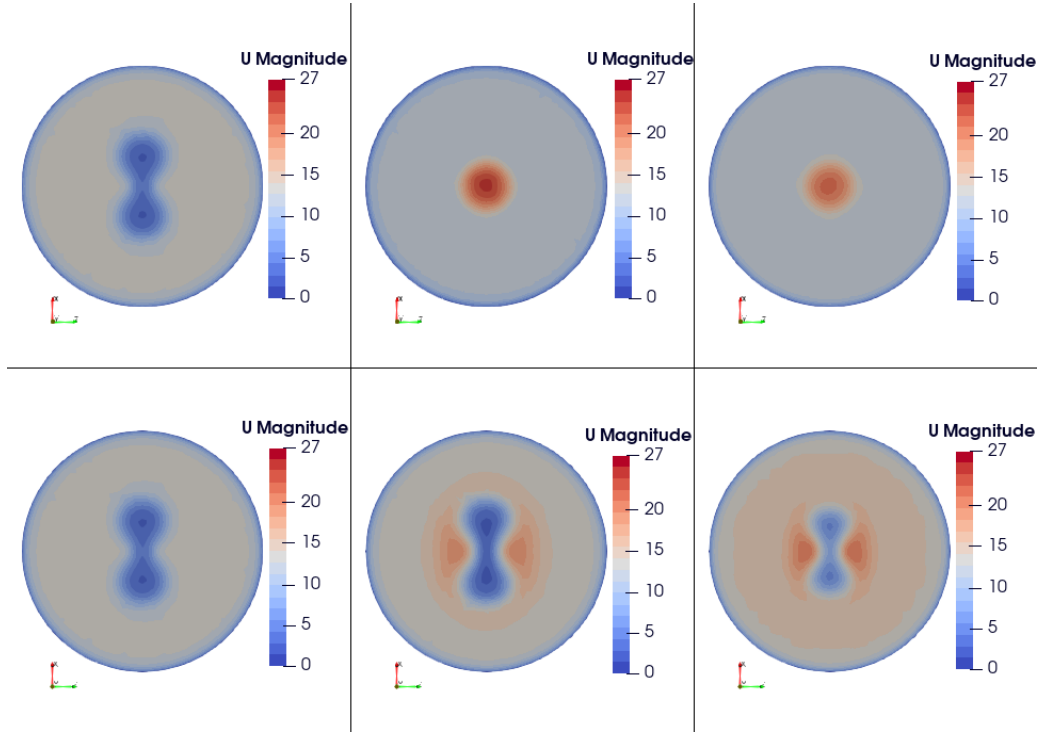


Figure 16: Velocity field distribution along y -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

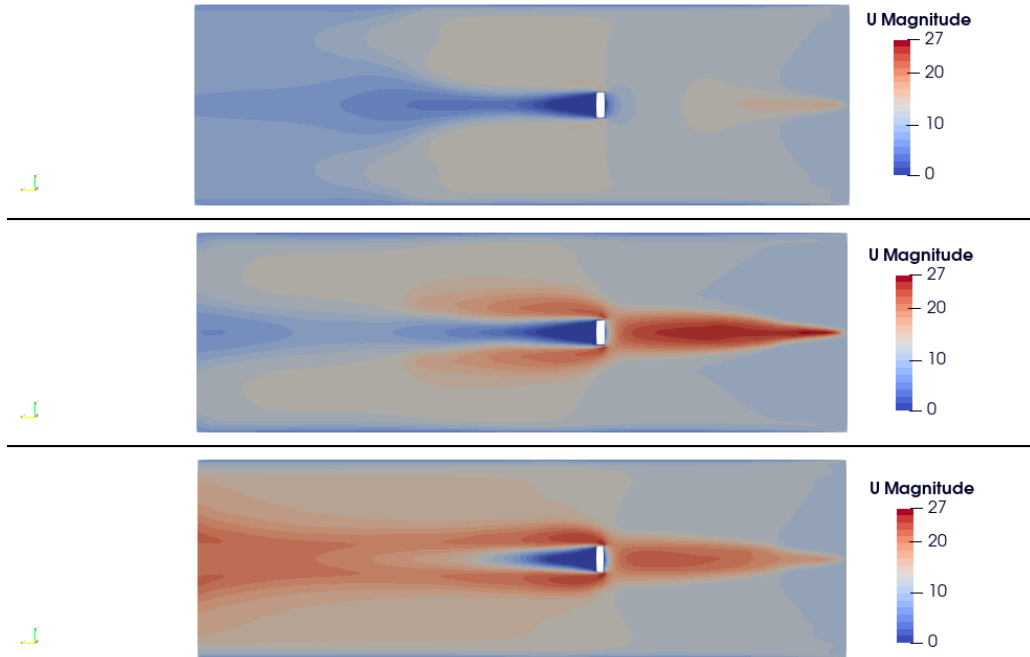


Figure 17: Velocity field distribution along x -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

Figure 18 shows the Mach number evolution in the domain. Its behavior is consistent with the one described above for the velocity field. throughout all the simulation the flow remains constantly subsonic.

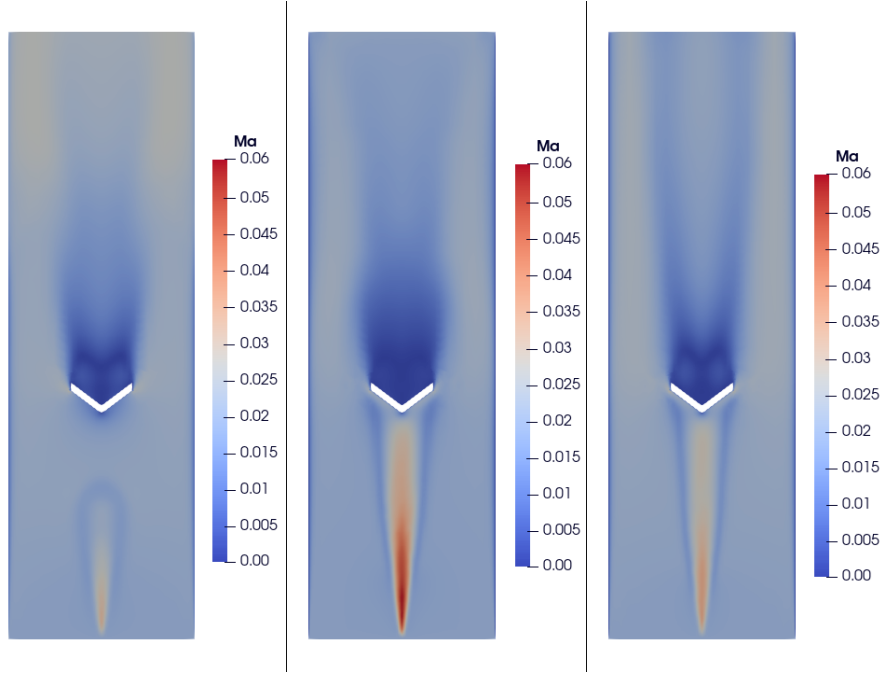


Figure 18: Mach field distribution along z -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

3.2.6 Vorticity Field Distribution

The vorticity evolution during the simulation highlights the presence of rotational flow regions near the splitter. In the initial stages, high vorticity is generated by the shear layers formed as the flow interacts with the V-shaped geometry. Over time, as the velocity field stabilizes, vorticity becomes more localized and concentrated along the edges of the splitter. As shown in Figure 19, this behavior reflects the development of coherent recirculation zones that contribute to mixing and flow symmetry downstream.

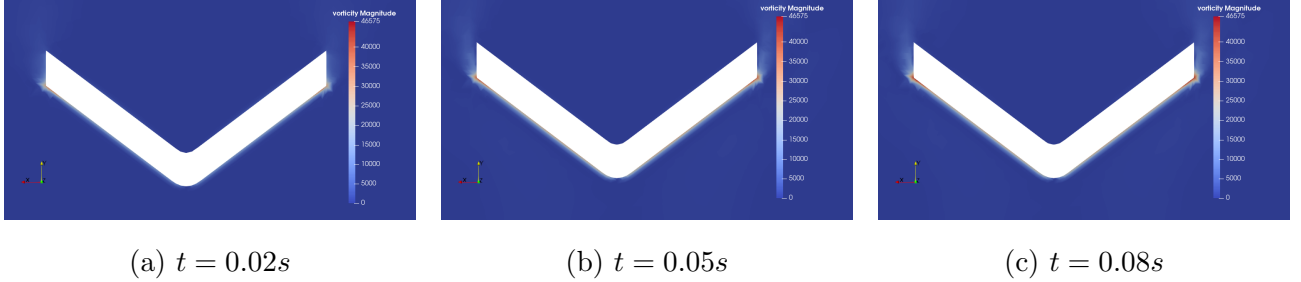


Figure 19: Vorticity field distribution along z -plane

3.2.7 Fuel Mass Fraction Distribution

Initially, the fuel mass fraction increases in the central region of the combustor as a result of continuous injection, as shown in Figure 20, Figure 21 and Figure 22. During the early stages of combustion, this region remains rich in fuel, providing the necessary conditions for ignition and flame stabilization. As combustion progresses, the fuel is progressively consumed, and by $t = 0.08s$, a clear reduction in concentration is observed.

In addition to the consumption, signs of radial diffusion become evident, especially near the splitter. These are primarily driven by the recirculation zones induced by the V-shaped geometry, which promote transverse mixing. The combined effect of fuel consumption and enhanced turbulent diffusion leads to a broader and more uniform distribution of the fuel fraction across the domain, contributing to a more stable and efficient combustion process.

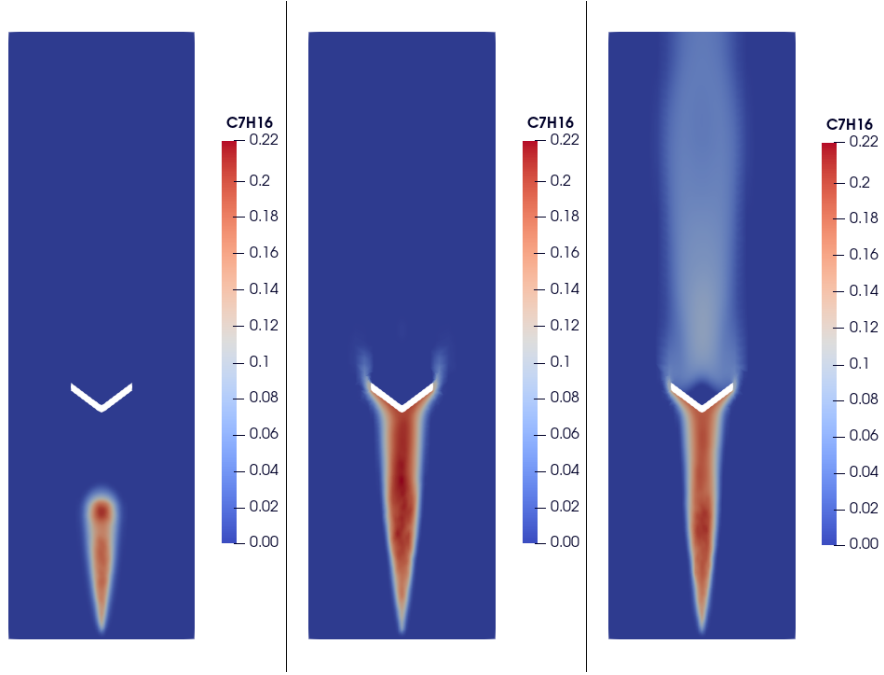


Figure 20: Fuel field distribution along z -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

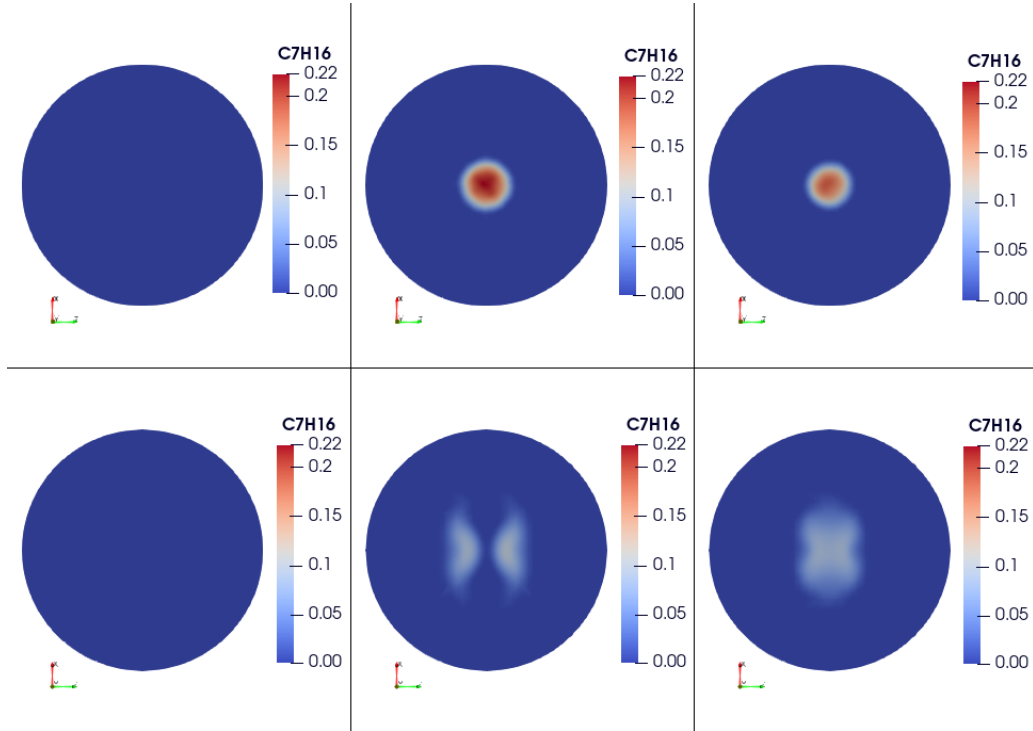


Figure 21: Fuel field distribution along y -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

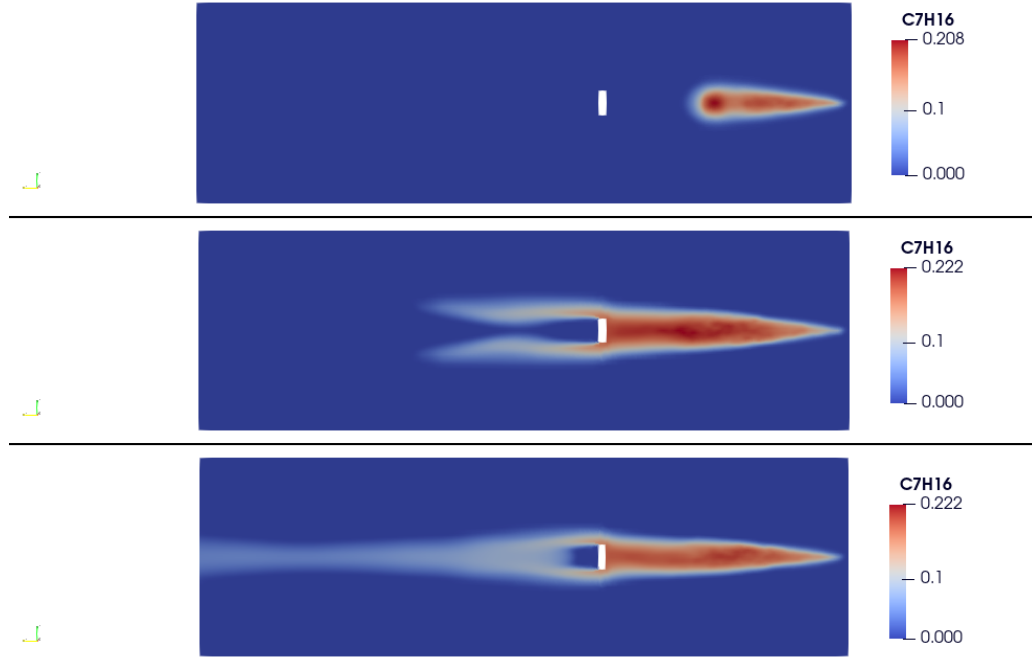


Figure 22: Fuel field distribution along x -plane at $t = 0.02s, t = 0.05s, t = 0.08s$ respectively

3.2.8 Isosurface at stoichiometric A/F ratio

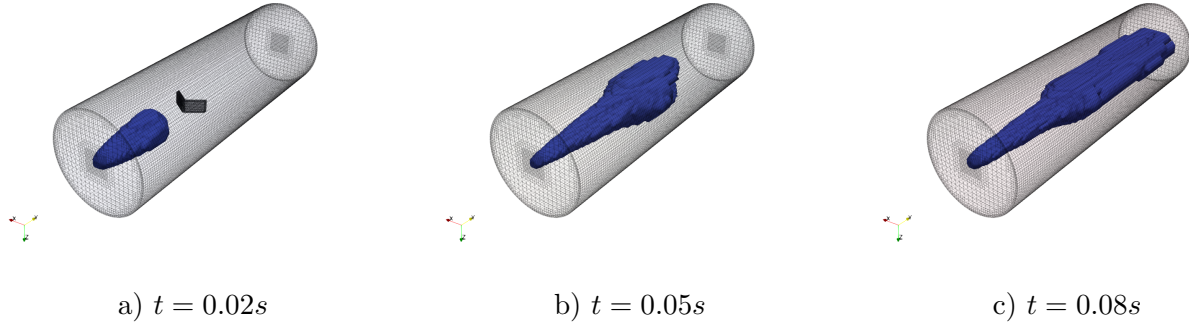


Figure 23: Isosurface of the air-to-fuel (A/F) molar ratio equal to 14.7, corresponding to stoichiometric conditions for air/ C_7H_{16} combustion.

The stoichiometric A/F ratio is a crucial parameter to understand how the combustion process is behaving. In fact, where the stoichiometric condition is reached we have also the maximum energy released by the combustion process, as all the reactants are consumed. Looking at the isosurfaces reported in Figure 23, the consideration regarding the evolution of the combustion process done in the previous section are confirmed. Here the stoichiometric A/F ratio for n-Heptane is assumed to be 14.7. In the early stage of the combustion, the profile where stoichiometric condition is reached is close to the injection point. In the following time-steps, the isosurface is enlarging and distributing more in the axial direction, meaning that the mixing is enhanced and the combustion process is becoming more effective, due to a more intense and stable flame.

3.2.9 Spray Distribution

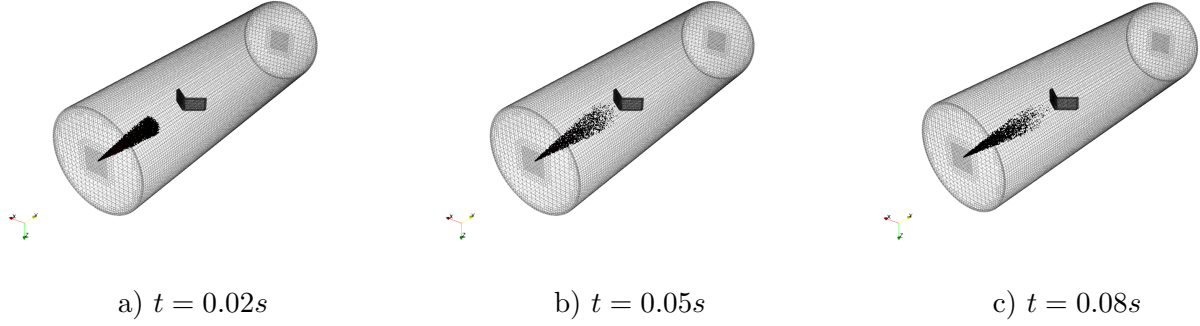


Figure 24: Spray particles evolution over time

The evolution of the spray particles over time is shown in Figure 24. At $t = 0.02$ s, the spray is still in its early phase, and the particle cloud appears dense and concentrated near the injector. This reflects the rising phase of the injection profile.

At $t = 0.05$ s, although the number of particles close to the source is reduced, the spray has propagated further downstream, reaching its maximum penetration. The increased momentum due to the peak mass flow rate allows the cloud to stretch longitudinally, despite a slightly lower local concentration. By $t = 0.08$ s, most of the spray has dissipated, and only a limited number of droplets remain in the domain.

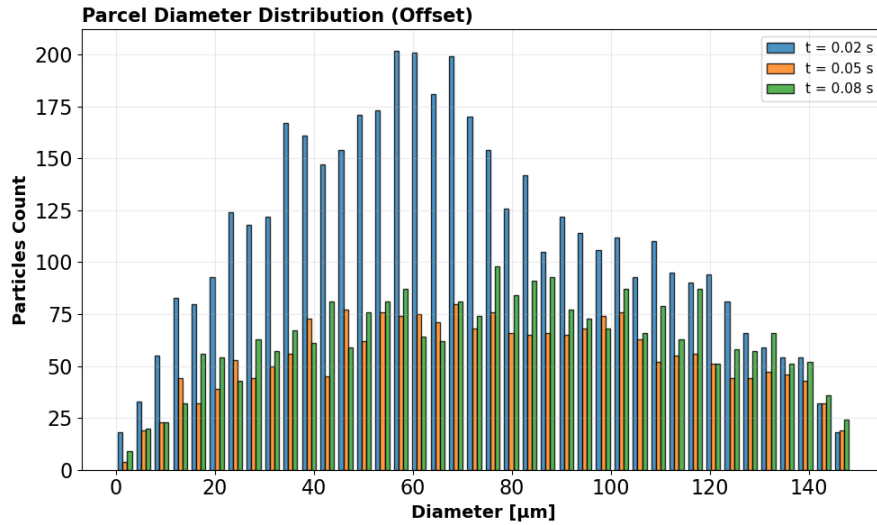


Figure 25: Particle size distribution over time

The particle diameter distribution shown in Figure 25 confirms the progressive reduction in particle count throughout the simulation. At $t = 0.02$ s, the number of parcels is significantly higher across all diameter ranges, especially between $40 \mu\text{m}$ and $90 \mu\text{m}$. As time progresses, both the number of droplets and their size decrease.

This behavior is mainly attributed to evaporation and turbulent dispersion. Smaller droplets tend to vaporize more quickly due to their higher surface-area-to-volume ratio, which explains the pronounced drop in the lower diameter classes. Meanwhile, larger particles are gradually transported downstream or broken up, contributing to the overall decrease in spray density observed at later time steps.

3.2.10 Species' Mass Fractions Evolution

In this section, the evolution of the mass fraction of each species involved in the combustion process—both reactants and products—is presented. For each species, values have been extracted along two different transverse lines: **line1**, located at $y = -90$ mm and **line2** at $y = 110$ mm.

These two sections have been selected as they lie upstream and downstream of the splitter, respectively. This choice allows for the evaluation of both the mixing effects resulting from the injection profile and the additional mixing promoted by the presence of the V-shaped splitter.

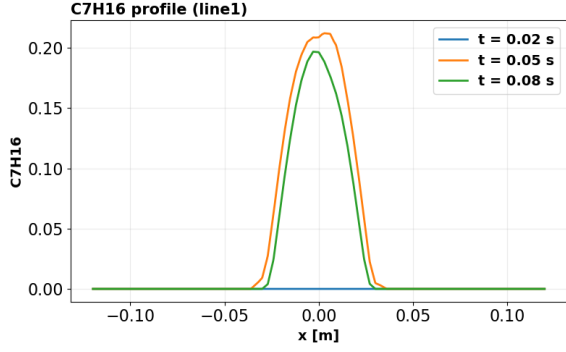


Figure 26: C_7H_{16} mass fraction evolution along **line1**

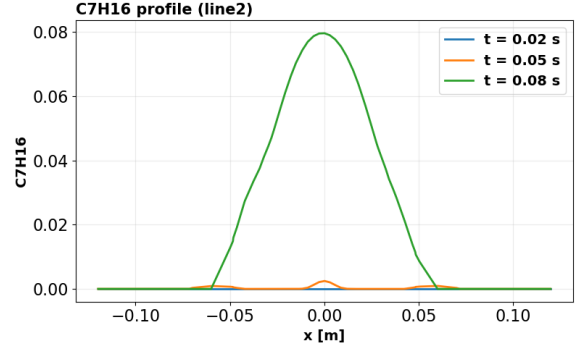


Figure 27: C_7H_{16} mass fraction evolution along **line2**

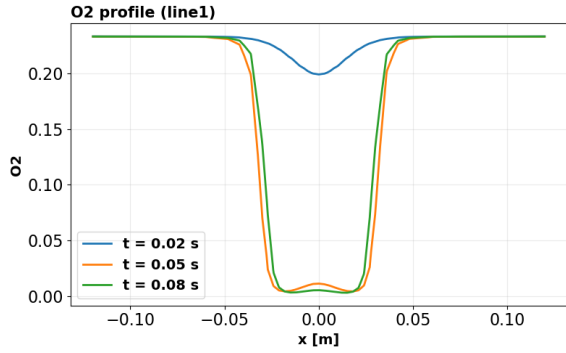


Figure 28: O_2 mass fraction evolution along **line1**

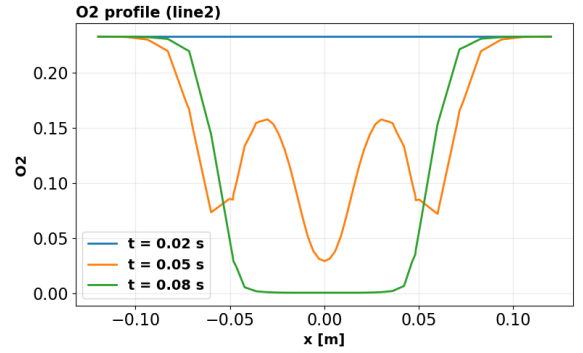


Figure 29: O_2 mass fraction evolution along **line2**

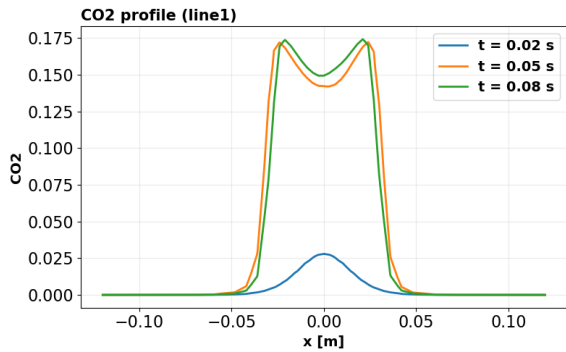


Figure 30: CO_2 mass fraction evolution along **line1**

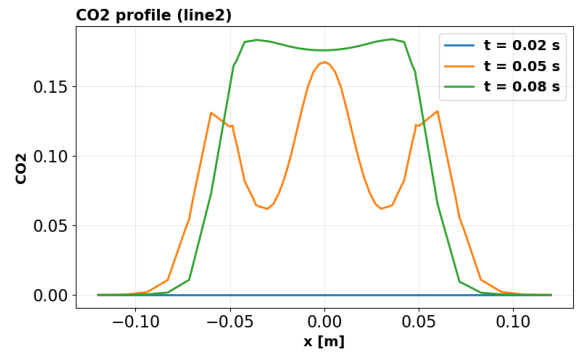


Figure 31: CO_2 mass fraction evolution along **line2**

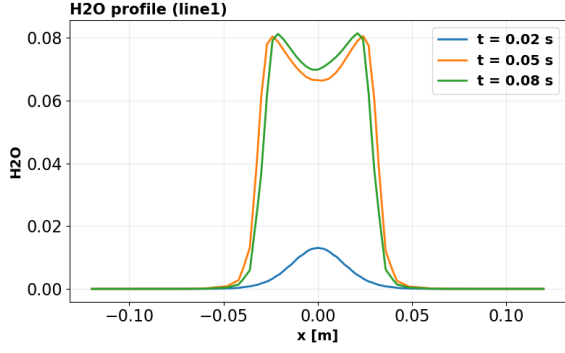


Figure 32: H_2O mass fraction evolution along line1

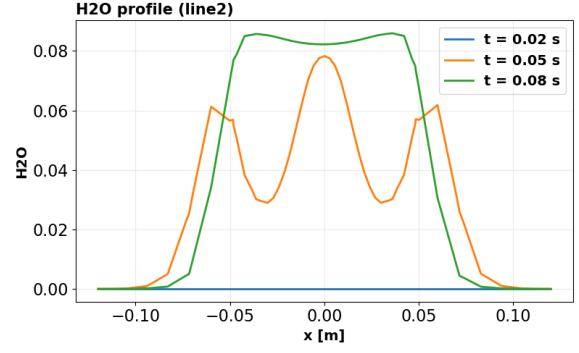


Figure 33: H_2O mass fraction evolution along line2

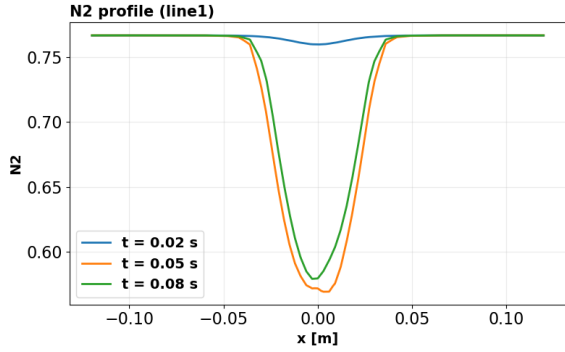


Figure 34: N_2 mass fraction evolution along line1

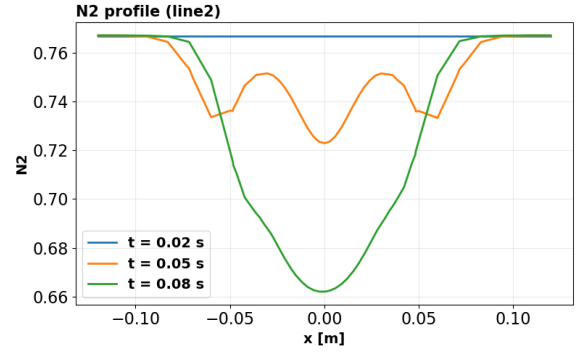


Figure 35: N_2 mass fraction evolution along line2

The temporal evolution of the chemical species along **line1** and **line2** shows how the combustion process develops inside the combustor. The mass fractions of the reactants, C_7H_{16} and O_2 , decrease over time, while the products CO_2 and H_2O increase. This confirms that combustion is taking place, with fuel and oxidizer being consumed and converted into stable products. The peak values of CO_2 and H_2O also move slightly downstream, showing that the reaction front is advancing as the flow develops.

The O_2 profile follows a similar trend to the fuel, confirming its role as the oxidizer. The N_2 profile stays nearly constant, as expected for an inert gas. When comparing the results from **line1** (before the splitter) and **line2** (after the splitter), it is clear that the species are more spread out and mixed after the splitter. This effect is stronger for the product species, which show smoother and more uniform profiles downstream.

These results highlight the role of the V-shaped splitter in improving the mixing of the flow and helping to stabilize the flame. The more uniform and symmetric profiles after the splitter indicate a transition from a localized combustion region near the injector to a more distributed and mixed regime further downstream. Overall, the species distributions are in line with the expected behavior of a reacting flow and confirm the influence of geometry on the combustion process.

3.2.11 Average temperature and pressure on inlet and outlet patch

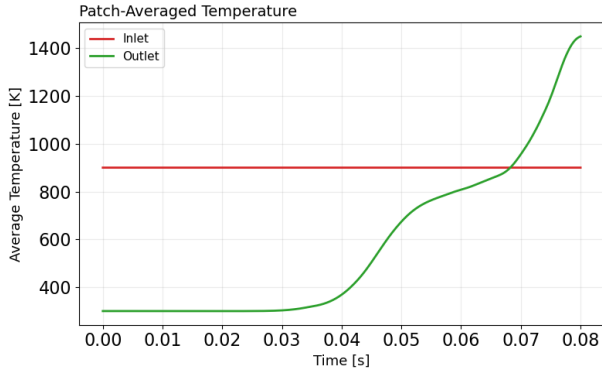


Figure 36: Average Temperature at inlet and outlet patches

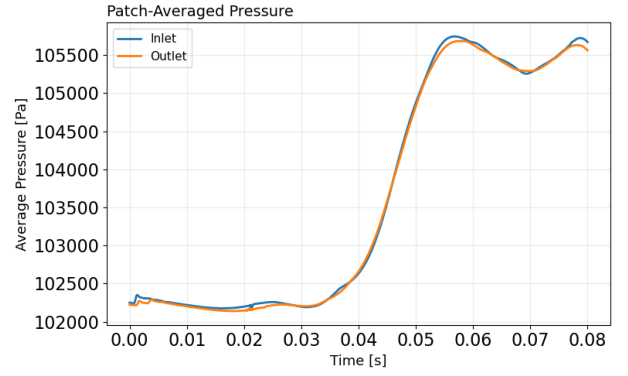


Figure 37: Average Pressure at inlet and outlet patches

The average temperature shown in Figure 36 is consistent with the imposed boundary conditions. At the inlet, the temperature is fixed at $T = 900$ K, and the plot displays a stable value over time, as expected. At the outlet, a `zeroGradient` condition is applied, allowing the temperature to evolve according to the local flow field. As combustion intensifies, the outlet temperature increases over time, in agreement with the heat release trends observed earlier.

Similar considerations apply to the average pressure evolution shown in Figure 37. Both inlet and outlet pressure follow a comparable trend, which is expected given the use of the `waveTransmissive` boundary condition. This condition minimizes spurious wave reflections and promotes a smooth transition between the inlet and outlet, thereby preserving the physical consistency of the solution throughout the simulation.